

Theorem (Cauchy–Schwarz). For any real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right).$$

1. Construction of a non-negative quadratic. Define

$$f(t) = \sum_{i=1}^n (a_i - t b_i)^2, \quad t \in \mathbb{R}.$$

Each summand is a square, hence $f(t) \geq 0$ for every real t .

2. Algebraic expansion. Using $(x - y)^2 = x^2 - 2xy + y^2$ and linearity of the sum,

$$\begin{aligned} f(t) &= \sum_{i=1}^n (a_i^2 - 2t a_i b_i + t^2 b_i^2) \\ &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2. \end{aligned}$$

Thus

$$f(t) = \left(\sum_{i=1}^n b_i^2\right) t^2 - 2 \left(\sum_{i=1}^n a_i b_i\right) t + \sum_{i=1}^n a_i^2. \quad (1)$$

3. Applying the discriminant condition. The polynomial in (1) has leading coefficient $A = \sum_{i=1}^n b_i^2 \geq 0$. Since $f(t) \geq 0$ for all $t \in \mathbb{R}$, its discriminant must be non-positive:

$$\Delta = \left(-2 \sum_{i=1}^n a_i b_i\right)^2 - 4 \left(\sum_{i=1}^n b_i^2\right) \left(\sum_{i=1}^n a_i^2\right) = 4 \left[\left(\sum_{i=1}^n a_i b_i\right)^2 - \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right) \right].$$

Thus $\Delta \leq 0$ is equivalent to

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right). \quad (2)$$

4. Degenerate cases. If $\sum_{i=1}^n a_i^2 = 0$ then each $a_i = 0$ and both sides of (2) equal 0. If $\sum_{i=1}^n b_i^2 = 0$ then each $b_i = 0$ and again (2) reduces to $0 \leq 0$. In either situation the quadratic $f(t)$ becomes a constant non-negative function, so the argument remains valid.

5. Equality condition. Equality in (2) occurs exactly when $\Delta = 0$. For a quadratic with non-negative leading coefficient, $\Delta = 0$ means there is a single real root λ such that

$$f(\lambda) = \sum_{i=1}^n (a_i - \lambda b_i)^2 = 0.$$

A sum of squares vanishes only when each square is zero, i.e. $a_i - \lambda b_i = 0$ for all i . Hence

$$a_i = \lambda b_i \quad (i = 1, \dots, n). \quad (3)$$

Conversely, if (3) holds for some $\lambda \in \mathbb{R}$ (including $\lambda = 0$ when one vector is the zero vector), then

$$\sum_i a_i b_i = \lambda \sum_i b_i^2, \quad \sum_i a_i^2 = \lambda^2 \sum_i b_i^2,$$

and substitution into (2) yields equality. Thus (3) characterises precisely the case of equality: the two vectors are linearly dependent.

□